

Mimosa Reproduction

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1. Reproducing the paper

Mimosa (Bi et al., Nucleic Acids Research 52:11455, 2024). Released model training/model_mimosa.pth on the paper's test sets data/miRAW_Test0-9.txt (548 positive + 548 negative pairs each; the 548 negatives are identical in all 10 sets), step size 5. Logs: logs/paper_reproduction/.

	Accuracy	PPV	Recall	Specificity	F1	NPV
Paper, Figure 3B (mean of 10 sets)	0.7568	0.6771	0.9332	0.5803	0.7932	0.8971
Our run (mean of 10 sets)	0.7567	0.6897 (AP 0.6771)	0.9332	0.5803	0.7932	0.8972

The original code computes "PPV" with average_precision_score (AP) instead of precision. The paper's 0.6771 equals the AP implied by its own recall and specificity; the true precision is 0.690. We report both from here on.

2. Changes to the original code

All experiments keep Mimosa's 40-nt architecture. Only training/Mimosa.py was modified:

Change	Why	Effect on the method
Command-line interface (--mode, --data, --stepsize, --threads), progress output, model loaded next to the script on CPU	Running the scripts on the cluster	None
PPV = precision_score; AP printed as "paper PPV"	Original PPV was AP (Section 1)	Reported metric only
Authors' commented-out torch.save line enabled	Without it training saves no model; the paper selects the minimum-validation-loss model	None (restores the documented behaviour)
--epochs argument, default 40	Needed for the 120-epoch run	None at the default

Not changed: architecture (16 encoder blocks, 8 heads, dim 64), input encoding, base-pairing features (Smith-Waterman of miRNA nt 1-10 against the 30-nt core), loss (CrossEntropy on softmax output, as in the original), Adam lr 1e-4 / weight decay 1e-5, batch 256, dropout 0.1, test procedure.

New files

Path	Purpose
training/prepare_dataset.py	Builds our dataset files (Section 3)
training/predict_site40.py	Gene-level test on our files by calling the authors' get_cts / kmers_predict / decision_for_whole; verified identical to the original perform_test on 100/100 miRAW Test0 pairs
scripts/run_prepare.slurm, run_train_site40.slurm, run_predict_site40.slurm, run_mimosa_all.slurm	Slurm jobs
experiments/cross_eval, window_analysis, composition	Diagnostic experiments (Section 6)

3. Data: sources, preparation and use

3.1 Raw inputs (outside the repository)

Input	Content	Used for
~/training_chunks/	346 JSONL chunks, 34,560,002 rows: 17,280,000 RNA-protein, 11,987,208 miRNA-mRNA, the rest other RNA-RNA types. A resampled training stream built from data_with_negatives final_train (every miRNA-mRNA row is in final_train; each pair repeated 4.3× on average, up to 200×)	Training pairs (miRNA-mRNA rows only)
~/data_with_negatives/rna_rna/miRNA_mRNA/	final_test_unseen_{pair,source,target}.jsonl	Test splits
~/fasta_files.zip	rna.fa (1.85 GB; read directly from the zip, T→U)	All sequences

Split semantics, checked against final_train positives: **unseen_source** = miRNAs never seen in training (targets 86 % seen); **unseen_target** = targets never seen (miRNAs 92 % seen); **unseen_pair** = neither seen. No test pair occurs in training. Negatives are made by swapping the miRNA or the target of a positive pair, so the same miRNAs and targets appear with both labels.

3.2 Preparation (training/prepare_dataset.py)

Step	Rows
miRNA-mRNA rows in training_chunks	11,987,208
Unique (miRNA, target, label)	2,761,271
Dropped: miRNA sequence not in rna.fa (mostly mirXplain isomiR names)	78,345
Dropped: target not in rna.fa (mostly mirXplain chromosome coordinates, some ENSG ids)	51,556
Dropped: miRNA longer than 30 nt (precursors; Mimosa pads miRNAs to 30 nt)	30,147
Dropped: target with non-ACGU characters / shorter than 40 nt	1,751 / 333
Dropped: label conflicts / pairs also in a test split	0 / 0
Kept	2,599,139
Used for training: pairs whose target is already a 40-nt site (given as a sequence in the data, mostly competitor_deeptargetpro), split randomly into train / validation (7 %, seed 42, stratified by label)	39,770 = 36,986 (20,647 pos / 16,339 neg) / 2,784 (1,554 / 1,230)

Mimosa's input is a 40-nt site, so only these rows can be used for training without changing the method. They have the same layout as the paper's training sites (30-nt core + 5 nt flanks, 5'→3'; the seed match sits at positions 25–29 in both). Test splits are written in full, in their original row order, with a `mimosa_status` column marking rows Mimosa cannot process (sequence missing, miRNA over 30 nt, non-ACGU):

Split	Rows	Processable	%
test_unseen_pair	44,580	42,476	95.3
test_unseen_source	398,828	379,727	95.2
test_unseen_target	415,348	394,110	94.9

3.3 Where each prepared file is used (data/custom/)

File	Used in
site40_Train_Validation.txt	Training, both runs (same format as the paper's miRAW_Train_Validation.txt)
pairs_test_unseen_{pair,source,target}.tsv	Gene-level tests (Section 5) and experiment A
sequences.fa	Sequences for all of the above (168,646 sequences)
data/miRAW_*	Paper reproduction, experiment B and the analyses in Section 6

4. Training

Original `perform_train`, unchanged, on `site40_Train_Validation.txt` (the paper trains on 58,793 miRAW sites). Validation metrics are site-level.

Run	Time	Best epoch	Val loss	Acc	PPV	Recall	Spec	F1	NPV
40 epochs (paper)	2 h 48 min	40	0.467	0.843	0.844	0.882	0.793	0.862	0.841
120 epochs	8 h 54 min	113	0.454	0.855	0.845	0.907	0.790	0.875	0.870

Model files: `runs/site40/model_concat_39.pth` (40 epochs) and `runs/site40_long/model_concat_112.pth` (120 epochs; file numbers count epochs from 0).

The 40-epoch run was still improving at epoch 40; the 120-epoch run plateaued after ~80 epochs (train loss kept falling, validation loss flat). **The 120-epoch model was used for all tests.**

5. Gene-level test

Procedure exactly as in the paper: target reversed to 3'→5', cut into 40-nt windows with step 5 (`get_cts`), every window scored with the authors' features (`kmers_predict`), pair positive if any window scores above 0.5 (`decision_for_whole`).

Runtime-only change: `kmers_predict` is called inside `torch.no_grad()`, which cuts memory from 4.4 GB to 0.94 GB per task on 4,076-nt targets; predictions are identical (checked on 100/100 pairs).

Split	Rows	n	Acc	PPV	AP	Recall	Spec	F1	NPV
test_unseen_pair	processable	42,476	0.514	0.511	0.510	0.881	0.141	0.647	0.537
	all rows	44,580	0.518	0.511	0.509	0.847	0.189	0.637	0.552
test_unseen_source	processable	379,727	0.516	0.512	0.511	0.881	0.143	0.648	0.540
	all rows	398,828	0.520	0.512	0.510	0.847	0.193	0.638	0.558
test_unseen_target	processable	394,110	0.512	0.509	0.508	0.882	0.136	0.645	0.530
	all rows	415,348	0.515	0.509	0.508	0.843	0.186	0.635	0.543
Paper, miRAW	10 × 1,096		0.757	0.690	0.677	0.933	0.580	0.793	0.897

"all rows": unprocessable pairs counted as negative predictions; they are mostly negatives, which raises specificity slightly. Results: `results/site40_long/<split>/pred_metrics.json` (metrics) and `pred.*.tsv` (one row per pair).

6. Diagnostic experiments

All use the original test procedure. Code and inputs in `experiments/`, outputs in `results/experiments/`.

6.1 Effect of target length (test_unseen_pair, 120-epoch model)

Target length (kb)	<0.5	0.5-1	1-1.5	1.5-2	2-2.5	2.5-3	3-3.5	3.5-4	≥4
Negatives called positive	0.25	0.77	0.84	0.87	0.86	0.88	0.88	0.89	0.91

False positives rise with the number of windows. Length alone does not explain the gap to the paper, though: our targets have 505 windows on average (median 2,612 nt), the miRAW test targets 408 (median 1,409 nt).

6.2 Cross-evaluation

Test data	Model	Acc	PPV	Recall	Spec	F1	NPV
A: test_unseen_pair, 4,000 random pairs (2,000 per label)	Authors' model	0.499	0.499	0.907	0.090	0.644	0.493
	Our model	0.521	0.512	0.889	0.152	0.650	0.580
B: the paper's 10 miRAW test sets (mean)	Authors' model	0.757	0.690	0.933	0.580	0.793	0.897
	Our model	0.513	0.507	0.894	0.131	0.647	0.556

Both problems are present: the authors' model fails on our data, and our model fails on the paper's data.

6.3 Window-level behaviour (100 negative + 100 positive pairs per dataset)

Data	Pairs	Model	Windows > 0.5	Positive windows per transcript (median)	Called positive
miRAW	negative	Authors'	13 %	0	39 %
		Ours	18 %	30	89 %
	positive	Authors'	63 %	154	95 %
		Ours	18 %	22	91 %
Ours	negative	Authors' / ours	60 % / 11 %	294 / 7	88 % / 85 %
	positive	Authors' / ours	59 % / 10 %	282 / 11	89 % / 93 %

Our model fires on a steady ~18 % of windows whatever the pair, so nearly every transcript gets at least one positive window. The authors' model fires on 63 % of all windows of miRAW positives, far more than the few binding sites a 3'UTR contains, i.e. it reacts to something other than individual sites.

6.4 What predicts the miRAW label without the model?

Feature (5-fold logistic regression or direct score)	miRAW ROC-AUC	Ours ROC-AUC
Target length	0.48	0.51
Target length + GC content	0.55	0.51
Target length + mono- and dinucleotide frequencies	0.53	0.50
miRNA alone: its positive rate in the paper's training data	0.86	-

The target transcript carries no label signal, but the miRNA does: miRNAs of miRAW test positives were positive in 84 % of their training examples, miRNAs of test negatives in 54 % (761 vs 116 distinct miRNAs; 5,379 vs 548 pairs).

6.5 miRNA swap on miRAW (150 targets per group)

Target	miRNA	Authors' model: called positive	Our model
From a negative pair	its own	0.413	0.873
From a negative pair	taken from a positive pair	0.947	0.933
From a positive pair	its own	0.933	0.873
From a positive pair	taken from a negative pair	0.387	0.873

The authors' model's decision follows the miRNA: a negative target becomes positive at the rate of real positives once it gets a "positive" miRNA, and vice versa. The published performance therefore rests largely on a miRNA-identity bias of the miRAW benchmark, which our benchmark does not have.

7. Deviations from the paper

- Training data: 39,770 40-nt sites from our dataset instead of 58,793 miRAW sites.
- Epochs: the tested model was trained for 120 epochs (the 40-epoch model was also trained; Section 4).
- Validation split: random 7 % of our training rows (the paper follows TargetNet's split).
- Test data: our unseen splits (several species, targets are transcript 3' ends up to 4,076 nt, swap negatives) instead of miRAW (human 3'UTRs). About 5 % of test pairs cannot be processed and are reported both ways.

8. Conclusions

- Mimosa trained on our data with the paper's method does not generalise to gene-level prediction (accuracy ~0.52 on all three splits, 0.51 even on the paper's own test sets).
- The published model is also at chance on our benchmark (accuracy 0.499).
- On miRAW the published model's decisions are driven by miRNA identity, which the miRAW benchmark leaks (ROC-AUC 0.86 from the miRNA alone). Our benchmark, where each miRNA and target appears with both labels, removes this shortcut; the published results do not transfer to it.